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RESEARCH-ARTICLE

Using Virtual Reality and Head-Mounted Displays to Increase Performance in Rowing Workouts

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ABSTRACT

Technology is advancing rapidly in the domain of virtual reality, as well as in using sensors to gather feedback from our body and the environment we are interacting in. Combining these two technologies gives us the opportunity to create personalized and reactive immersive environments. These environments can be used for training of dangerous situations (e.g. fire, crashes, etc) or to improve skills with less distraction than regular natural environments would offer. The pilot study described in this paper, puts an athlete rowing on a stationary rowing machine into a virtual environment. The virtual reality receives movement data from several sensors of the rowing machine and displays those in the head-mounted display. In addition, metrics on technique are derived from the sensor data as well as physiological data. All this is used to investigate if and to which extend VR improves the technical skills of an athlete, during the complex sport of rowing. Furthermore, athletes are given subjective feedback about their performance comparing the standard rowing workout with the workout using VR. First results indicate improved performance of the workout and an enhanced experience for the athlete in the VR condition.

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1 INTRODUCTION

The evolution of videogames has always focused on the task of offering a more immersive experience, with better graphics, better sound, better interaction, etc for the user. With the presentation of Nintendo Wii, a new stage of videogames was created. The physical activity while playing sport games was the added feature, and thus, also the level of immersion in sports videogames was enhanced for the users. It allowed the user to play with movement controllers and it was the first console which brought the sports field to the living room. After this, the Kinect appeared on the market which is

able to detect the body movements and interprets those movements as input signals to control the videogame. Next, head-mounted displays (HMD) using virtual reality (VR) entered the market. These devices bring the user the opportunity to get a 360-degree experience of the game they are playing. VR is also starting to be used during sports workouts both by professionals and amateurs. On the one hand, it gives the athlete the opportunity to replay certain scenes of previous games. This can be especially interesting for team sports (such as football, basketball, etc.). On the other hand, it offers the possibility to perform entire training-sets in VR. This can be used, in order to make monotonous indoor-workouts more fun, or preparing the athlete for how the competition arena will look like and let them perform there.

In the domain of sports and VR, so-called exergames put the focus on the gaming engine and thus rather have the goal to motivate people to exercise. The fun and motivating part is more important than the coaching on perfect technique or to perform optimal [2]. In addition, many approaches in VR sports try to simulate realistic sports environment and are putting the focus on improving technique and add more realistic conditions for the athlete. However, improving the fun during workout is also an important aspect here.

Appelbaum et al. [1] gives a review of sports vision training using digital training techniques which is concentrating on team-sports and fast-paced sports. Mentioning that athletes rely greatly on vision, and visual training for their sports such that they can improve their performance using VR. In the review, three naturalistic sports training approaches are highlighted, one of them being simulations to recreate the sporting environment in virtual reality. While this is a growing market especially for amateurs, only very limited research has been performed in this area which can show an increase in performance.

Using virtual reality during monotonous indoor endurance workouts has become very popular with the rise of Zwift¹. Giving cyclist and runners the opportunity to perform their workouts on stationary indoor bikes or treadmills online, while following a virtual course and race against other athletes worldwide.

Neumann et al. [8] give an overview on using virtual reality technology in endurance sports. Their conceptual framework for applications in VR sports consists of four major factors: Task, User, VR Environment, and Non-VR environmental factors. In their review, they highlight that most research so far has only been concentrating on the task, user, and VR-environment factors. The most investigated task was cycling; running and rowing have only been subject to a few studies. None of the studies they present used

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¹www.zwift.com

HMD as the display for the virtual environment. Furthermore, it gets obvious that most studies rather use novice users to the sports, as compared to experienced.

Very recent work from Shepherd et al. [11] uses HMD while athletes cycle in a virtual velodrome. The non-elite cyclists evaluate their experience during cycling in the VR. The study results showed an increase in behavioural fidelity as well as in performance which is including the concept of presence. The studies of [10] and [5] used rowing and head-mounted displays. Where [10] investigated the impact of a virtual environment in a single-player rowing exergame. As virtual environments a CAVE environment was compared to a HMD. Different feedback mechanisms to motivate people to row at a certain pace. And [5] shows how VR and the opportunity to talk to the opponent influences the performance of the athlete in a multi-player exergame.

Previous studies have shown that bringing athletes into a virtual reality scenario, their performance in terms of covered distance / power output increases. This was shown in different endurance sports such as rowing [3], running on a treadmill [9], or cycling [7]. All these three studies and most studies in the current literature are focussing on performance improvement (based on time, power, etc) and/or increase of mood and not on improving technique using VR systems.

Only little work has been done on the focus on comparing technique when work out in standard mode versus virtual reality. Sigrist et al. [12] investigated how different feedback mechanisms and the congruency of them can improve technique when rowing on a trunk-arm. Visual feedback was given in the sense of showing the trajectory of an optimal stroke and the comparison of the participants stroke. In addition, auditory and haptic feedback was evaluated. The results suggest that congruent audiovisual feedback provides the best feedback for learning of the movement. The visual feedback in this study was projected on the wall.

In this paper, we present a study within indoor-rowing. Where we explore how the VR environment presented in a head-mounted display unit is affecting the performance and the experience of athletes over a regular indoor-workout.

2 METHODS

2.1 Hardware

In the experiment, we used a rowing machine, Augletics Eight2, from Augletics². The ergo-meter is equipped with several sensors which in addition to standard rowing machines is capturing movement of the sledge and the handlebars. Furthermore, the machine provides with information about the power on each stroke, duration of the stroke, recovery slide, strokes per minute, etc. Using these objective measures, different metrics on rowing technique are being derived. Feedback on these metrics is given on the display of the rowing machine itself 1b. The derived metrics are consistency, movement, stroke length, recovery, rhythm (see Table 1). Each metric reaches values between 1 and 100, where 1 is very poor performance and 100 perfect.

An API provides parameters such as stroke force and speed via an HTTP connection. Two oars and a scull are modelled by a linear

Metric	Description
Stroke length	Stretch further! Roll forward until your shins are in a vertical position
Recovery	Try to roll forward slowly and steadily. This put less strain on your joints
Rhythm	Pull handle quickly towards your chest then roll slowly forward. A good rhythm is 20 strokes per minute
Consistency	Try to make every stroke like the last one, using the same amount of strength and the same technique
Movement	Extend your arms first, then move your upper body forward and only then begin to bend your knees and roll forward

Table 1: Metrics from the ergo-meter provided by Augletics.

interpolation between the ergometer parameters and the game object positions. The resistance of the ergometer was set to a medium value (5 out of 10), to avoid rapid fatigue of users. Unity project was used for the visualization of the virtual environment (a summer lake surrounded by distant mountains and a red scull). Very realistic haptic feedback was provided by using the ergo handles in the real world. For displaying the virtual world scenario, a HTC Vive set was used. Participants were wearing the headset during the workout. The controllers were placed next to the rowing machine, to indicate to the VR system the location and orientation of the ergometer. The controllers were not moved during the sessions.

In addition to the sensors on the machine, the participants wore a belt around their chest to measure the breathing rate during exercise. The breathing rate from the Sweetzpot³ breathing sensor was read every 50 m.

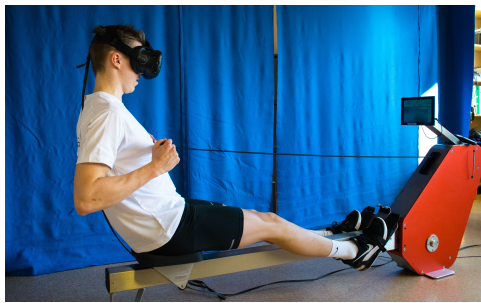
2.2 Experimental Design

Participants filled in a demographic questionnaire in the beginning. Afterwards, they performed a warm-up/baseline phase, in which they were rowing 500m, in both, the non-VR and VR setup, in order to get accustomed to the rowing machine and the VR environment. During the non-VR setup, participants saw all the available feedback including the metrics regarding their technique. In the VR-case, the only feedback they received was passing a distance indicator-line every 50m. The warm-up session was followed by the two test conditions.

During the experiment, two conditions were tested. (1) A regular workout on the ergo-meter where participants only saw the distance covered as feedback during their workout. (2) A workout on the ergo-meter in the virtual environment. Here, participants were wearing the HTC Vive headset (Fig. 1a), and were put on an artificial lake to row (Fig. 1b). The feedback that was provided in the second scenario was distance markers every 50m besides the steady movement on the lake depending on the force applied to the oars. During both sessions, the participants needed to cover a distance of 500m rowing on the machine. The order of VR vs non-VR was

²www.augletics.de

³www.sweetzpot.com



(a) Rowers on ergo-meter using VR



(b) Screenshot of VR

Figure 1

randomized between participants. The breathing rate was noted every 50m, and was averaged afterwards over the session.

After each of the two sessions, participants were asked to fill in the Activity Flow Scale questionnaire, to assess their level of immersion, flow and satisfaction with the rowing experience. In addition, participants reported on their emotional state after the exercise on a 9-point-SAM [6] questionnaire, which contained the dimensions of valence, arousal, and dominance.

2.3 Questionnaires

After welcoming the participants, and let them read the instructions of the experiments, a demographic questionnaire was completed by each participant. Here, age and gender was retrieved first, followed by how many hours per week participants train on average and how often per week they exercise. Next, they answered how much experiences they have in rowing and in the use of VR technologies, such as HMD, Cave environment, etc.

After each test condition, participants filled out the Activity Flow State Scale (AFSS) questionnaire [4], with 24 items using four questions for each of the eight factors; which are: actions and awareness merging; clear goals; concentration on task at hand; unambiguous feedback; challenge-skill balance; transformation of time; sense of control; loss of self-conscious; autotelic experience. For analysis, the four questions for each of the factors were averaged. In addition to the AFSS, questions concerning the participants perceived level of motion-sickness, the quality of the overall experience with the test condition were asked for, and in case of the VR condition judging the visual quality. All questions were rated on a five-point Likert scale (Results were afterwards transferred as -2='Strongly Disagree', -1='Disagree', 0='Neutral', 1='Agree', 2='Strongly Agree').

Furthermore, a 9-point-SAM scale was deployed, after each condition asking for the level of valence, arousal, and dominance during each condition.

3 RESULTS

16 participants (5male, 11female) took part in the study with an average age of 23.4 years. 13 of them stated to be experienced rowers, while the other 3 were non-regular rowers. Except for one participant all subjects are very involved in training, and stated to train 9.4 hours per week in average.

3.1 Objective Data

Table 2 shows the results of the objective metrics provided by the ergo-meter manufacturer as well as the value for breath per minute, taken from the breathing sensor, during both conditions. The results show marginal improvements on most of the statistics.

The objective measures taken during the experiment and the metrics derived based on those, show that values are generally slightly higher for the VR scenario than for the non-VR scenario. With the exception for the breathing rate which is slightly smaller. The biggest difference being in *Rhythm*, followed by *Recovery*. A paired t-test comparing VR and non-VR condition for the collected data reveals a significant differences between the two scenarios (VR vs non-VR) for *consistency* ($p=0.1$), *recovery* ($p=0.09$), and *rhythm* ($r \leq 0.01$). Furthermore, the average breathing rate for the non-VR condition was slightly higher as for the VR case, also showing a significant difference ($r \leq 0.01$).

Correlations calculated between the different metrics show a strong correlation ($r \geq 0.7$) between *Breath* and *Strokes/Min*, *Breath* and *Rhythm*, *Stroke Length* and *Rhythm* and *Stroke Length* and *Movement* can be seen. As well as a moderate correlation ($r \geq 0.5$) between *Breath* and *Movement*, *Breath* and *Stroke Length*, *Breath* and *Recovery*, *Consistency* and *Time* and *Rhythm* and *Strokes/Min* (see Table 4).

3.2 Subjective Data

Subjective data was gathered after each session on the 24-item Activity-Flow-Scale questionnaire. Results of questions for each participants were averaged over the four questions for each category. This questionnaire was extended by questions towards the entire experience of the workout and in case for the VR-case by the evaluation of the visual quality.

Table 3 shows the averaged data from the questionnaire filled in after each of the conditions.

Transformation of Time (TT) gave the biggest difference between both conditions, showing that participants lost their sense of time more during the VR workout as compared to the standard workout. *Valence* ratings were higher for the VR case than the non-VR case, the same absolute difference was observed for the *dominance* ratings with the difference that participants felt less in control in the VR condition compared to the non-VR case. The *autotelic experience* was also rated significantly higher for the VR condition compared to the standard workout. People were reporting on a slightly elevated level of motion sickness for the VR case than for the non-VR case. However, no participant needed to quit the session nor was it reported to be uncomfortable.

	Breaths/min	Consistency	Movement	Stroke Length	Recovery	Rhythm	Time (s)	Strokes/min
VR	19.25	85.28	85.26	97.92	75.89	79.41	137.75	23.63
Non-VR	20.41	84.36	83.36	97.14	72.63	73.98	135.69	22.73
Delta	-1.15	0.92	1.89	0.78	3.26	5.44	2.06	0.90
T-Test	$\leq 0.01^{**}$	0.10*	0.10	0.40	0.09*	$\leq 0.01^{**}$	$\leq 0.01^{**}$	0.43

Table 2: Average over all subjects of recorded objective data for the VR and Non-VR condition. Delta is the difference between VR and Non-VR, T-Test shows the p-values of the T-Test between the two conditions. Consistency, Movement, Stroke Length, Recovery and Rhythm are metrics taken from the ergo-meter.

	SAM1	SAM2	SAM3	MAA	CG	CO	UF	TT	CN	SC	AE	MS	EX	QoE	VQ
VR	7.56	4.63	6.44	0.88	0.25	0.29	0.03	0.54	0.50	1.40	1.00	1.34	1.13	1.25	0.63
Non-VR	6.50	4.00	7.06	0.96	0.36	0.46	0.41	-0.17	0.66	1.33	0.42	1.78	0.88	0.81	0.00
Delta	1.06	0.63	-0.63	-0.08	-0.11	-0.17	-0.38	0.71	-0.16	0.06	0.58	-0.44	0.25	0.44	0.63
T-Test	$\leq 0.01^{**}$	0.14	0.15	0.62	0.27	0.01*	0.11	$\leq 0.01^{**}$	0.53	0.66	$\leq 0.01^{**}$	0.07*	0.10	0.09*	0.03*

Table 3: Average over all subjects of recorded subjective data for the VR and Non-VR condition. Delta is the difference between VR and Non-VR, T-Test shows the p-values of the T-Test between the two conditions. SAM1=valence; SAM2=arousal; SAM3=dominance; MAA=merging actions and awareness; CG=clear goals; CO=Concentration on task at hand; UF=unambiguous feedback; CS = challenge skill balance; TT=Transformation of time; CN=Sense of Control; SC=loss of self-conscious; AE = atotelic experience; MS=Motion sickness; EX=Experience; QoE=Quality of Experience; VQ = Visual Quality

	Breaths/min	Consistency	Movement	Stroke Length	Recovery	Rhythm	Time	Strokes/Min
Breath		-0.13	-0.61**	-0.51*	0.64***	-0.74***	-0.35	0.85***
Consistency	0.62		-0.10	-0.25	-0.02	-0.13	0.62**	-0.15
Movement	0.01**	0.71		0.81***	-0.39	0.87***	-0.22	-0.40
Stroke Length	0.05*	0.35	0.01***		-0.28	0.74***	-0.36	-0.40
Recovery	0.01***	0.95	0.14	0.3		-0.37	-0.24	0.36
Rhythm	0.01***	0.62	0.01***	0.01***	0.16		-0.18	-0.60**
Time	0.18	0.01**	0.42	0.17	0.37	0.5		-0.40
Strokes/Min	0.01***	0.58	0.12	0.12	0.17	0.01**	0.13	

Table 4: Results of Pearsons correlation between the different metrics. Lower left diagonal shows p-values of the correlation, upper right shows correlation values. ***indicates correlations with p values lower 0.01; ** correlations lower 0.05; *correlations lower than 0.1.

A T-Test was calculated comparing both conditions and yields significant results for 'Transformation of time' ($r \leq 0.01$), the valence scale by the SAM questionnaire ($r \leq 0.01$), 'Autotelic Experience' ($r \leq 0.01$), 'Concentration on Task-at-Hand' ($p=0.01$), 'Motion Sickness' ($p=0.07$), 'Quality of Experience' ($p=0.09$) were significant.

Correlations calculated for the subjective data shows strong correlation ($r \geq 0.7$) between *Valence* and *Autotelic experience*, and moderate correlations ($r \geq 0.5$) for *Dominance* and *Unambiguous feedback*, *Dominance* and *Sense of Control*, *Sense of Control* and *Unambiguous Feedback*, *Valence* and *Quality of Experience*, and *Quality of Experience* and *Autotelic experience*.

3.3 Combined Data

Investigating relationships between the subjective data and the objectively extracted metrics indicates only a few weak ($r \geq 0.3$) but significant ($r \leq 0.1$) correlations between the different data; such as *Unambiguous feedback* and *Stroke Length*, *Transformation of time*

and *Stroke Length*, *Arousal* and *Rhythm*, *Rhythm* and *Unambiguous feedback*, *Loss of self-conscious* and *Rhythm*, *Motion sickness* and *Strokes/Min*, and *Strokes/Min* and *Visual Quality*.

3.4 Qualitative Data

After the experiment some participants gave informal feedback of things to improve the system. This was not a planned part of the experiment, but contains helpful information for further development and research in this area. The main feedback here was that a multiplayer mode would be desirable. Here two different versions are of interest: race against others; and to row in a team. Especially the latter one is interesting for experienced rowers, as synchrony between the rowers in one boat is one of the main challenges in team rowing. Having the chance to row in a virtual team boat and work on the synchrony of the row would be a very beneficial training especially during winter.

Rowing against others in a multiplayer game would especially be fun and motivating when having avatars of the opponents. In

	SAM1	SAM2	SAM3	MAA	CG	CO	UF	TT	CN	SC	AE	MS	EX	QoE	VQ
SAM1		0.03	0.36**	-0.01	-0.01	-0.04	0.11	0.37**	0.09	-0.07	0.81***	0.17	0.08	0.70***	0.43**
SAM2	0.89		-0.17	-0.18	-0.05	0.27	-0.16	-0.21	-0.23	-0.20	-0.01	-0.45***	0.04	0.09	0.02
SAM3	0.04**	0.35		0.28	0.30*	-0.10	0.63***	-0.18	0.65***	-0.10	0.27	0.36	-0.09	0.38**	0.41**
MAA	0.97	0.33	0.13		-0.08	-0.30*	0.41**	0.06	0.44**	0.07	0.08	0.08	0.28	-0.16	0.08
CG	0.96	0.8	0.09*	0.67		0.30	0.00	-0.16	-0.01	-0.11	0.06	0.09	-0.11	0.02	-0.06
CO	0.84	0.13	0.59	0.09*	0.1		-0.31*	-0.22	-0.18	-0.06	0.05	0.00	-0.18	-0.02	-0.06
UF	0.56	0.39	0.01***	0.02**	0.98	0.08*		-0.20	0.62***	0.15	0.09	0.30*	-0.03	0.26	0.30
TT	0.04**	0.25	0.32	0.75	0.38	0.22	0.28		-0.21	0.18	0.41**	-0.10	0.17	0.07	0.01
CN	0.62	0.21	0.01***	0.01**	0.94	0.31	0.01***	0.25		0.04	0.20	0.31*	-0.13	0.05	0.43**
SC	0.69	0.27	0.57	0.7	0.55	0.75	0.42	0.34	0.84		0.13	0.11	0.39**	0.13	0.18
AE	0.01***	0.94	0.13	0.66	0.74	0.77	0.63	0.02**	0.28	0.48		0.09	0.21	0.63***	0.48***
MS	0.37	0.01***	0.04	0.65	0.63	0.98	0.1*	0.58	0.09*	0.55	0.62		-0.02	0.29	0.26
EX	0.68	0.84	0.63	0.13	0.56	0.33	0.86	0.36	0.48	0.03**	0.25	0.91		0.04	0.00
QoE	0.01***	0.62	0.03**	0.37	0.9	0.9	0.16	0.7	0.8	0.47	0.01***	0.11	0.83		0.46***
VQ	0.02**	0.91	0.02**	0.68	0.73	0.73	0.98	0.97	0.01**	0.32	0.01***	0.16	1	0.01***	

Table 5: Results of Pearsons correlation between the different subjective scores. Lower left diagonal shows p-values of the correlation, upper right shows correlation values. *** indicates correlations with p values lower 0.01; ** correlations lower 0.05; *correlations lower than 0.1.

	Breaths/min	Consistency	Movement	Stroke Length	Recovery	Rhythm	Average	Time	Strokes/Min
Q1	-0.09	0.19	-0.03	-0.17	0.20	0.18	0.16	0.28	0.11
Q2	-0.23	-0.12	0.34*	0.36*	-0.20	0.36**	0.31*	-0.24	-0.11
Q3	0.05	0.24	-0.06	-0.16	-0.13	-0.11	-0.16	0.18	0.12
MAA	0.12	0.17	0.05	-0.27	-0.20	-0.08	-0.14	0.23	0.20
CG	-0.04	0.07	0.03	0.02	-0.16	-0.02	-0.07	-0.05	0.08
CO	0.23	-0.10	0.07	0.13	0.24	0.02	0.19	-0.25	0.17
UF	0.28	0.14	-0.35*	-0.36**	-0.04	-0.37**	-0.43**	-0.09	0.29
TT	0.13	-0.07	-0.24	-0.40**	0.09	-0.06	-0.18	0.13	0.30*
CN	0.18	0.20	-0.20	-0.20	-0.07	-0.33*	-0.32*	0.10	0.35*
SC	0.12	0.02	-0.26	-0.26	0.17	-0.40**	-0.29	-0.17	0.15
AE	0.06	0.28	-0.07	-0.18	0.30*	0.05	0.13	-0.01	0.39**
MS	-0.11	-0.05	-0.14	-0.16	-0.14	-0.02	-0.19	0.19	-0.02
EX	-0.27	0.34*	0.21	-0.05	-0.09	0.24	0.18	0.09	-0.23
QoE	-0.18	0.01	0.10	0.03	0.00	0.23	0.16	-0.08	0.05
VQ	0.07	0.22	-0.16	-0.07	0.02	-0.21	-0.18	0.11	0.36**

Table 6: Results of Pearsons correlation between the different metrics. Table shows correlation values between the subjective scores and the objective metrics. *** indicates correlations with p values lower 0.01; ** correlations lower 0.05; *correlations lower than 0.1.

addition, participants were saying that otherwise they would prefer to have a simple red line following them, to keep a certain speed compared to more playful elements such as following sharks or similar. Having a plain speed indicator for experienced athletes is also in line with previous research [10].

Participants were mentioning that currently there is too little feedback in the VR condition. They suggested that the optimal place for feedback would be the virtual place between the legs. The same position where they also get feedback during outdoor rowing.

4 DISCUSSION AND CONCLUSION

The objective results above indicate that athletes perform slightly better with regard to technical aspects in the VR-condition compared to the traditional workout. Though also the performance data such as completion time show a significant difference. We will disregard this for the further analysis as participants were instructed to concentrate on technique rather than on time. Indicating that an improvement of time may be a by-product of an improved technique or just putting unknowingly more effort into one of the conditions.

Rhythm and *Breath* were the two metrics that improved the most in the VR case compared to the non-VR case. Also, the strong

correlation between these two metrics implies that VR helps to improve the rhythmic pattern which is essential when it comes to rowing.

When it comes to the objective metrics provided by the ergometer, it offers two different operating modes for the feedback, *normal* and *advanced*. In our study we used the *normal* mode. In case of experienced athletes this may not be the ideal setting. In future studies it would be best considering to change to *advanced* mode when performing the experiment with experienced rowers. In this mode, the metrics are more rigorous, which may be more appropriate for experienced rowers. This change may result in larger differences between the two conditions concerning the technique metrics.

On the subjective scores, participants reported on a different experience between the two scenarios. The scale *Concentration on the task* shows a higher score for the traditional workout. During this condition athletes obviously have fewer distraction and therefore can concentrate much more on the technical aspects of the workout. However, although participants report to be less concentrated on the task per-se, the technique metrics increased. Here, especially the ones related to the rhythmic movement exertion. The fact that participants enjoyed the VR experience more than the non-VR (valence score increased), also may be in line with them reporting to have a higher score on *loose sense of time*. Due to the distraction of the VR, they lost easier track of time. This could be an important point why people may use increasingly indoor workouts enriched by virtual environments. The reported emotional feedback from the questionnaires shows also an increase in the arousal score, but athletes felt less in control of the rowing experience (dominance) for the VR case than the traditional condition. They also scored higher for the autotelic experience.

From the results we can in general conclude that there is a difference when working out in VR versus a standard workout. This was only a first study to investigate possible effects. In this experiment, we concentrated rather on technique of the rower, and not so much on the best performance time-wise. Future studies could investigate how VR is affecting the rowers performance when working on more exhausting zones during the workout. Also, what effects in VR would be improving the rower and their experience? Working out in a more social context such as rowing versus friends or opponents could be interesting to investigate. Thus varying the non-VR environmental factors could give interesting insights. An other aspect is to investigate the different demands non-experienced rowers vs experienced rowers would have on such a VR rowing system.

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